Internet Explorer

Assignment Date: 09/02/2026
Assignment Name: Internet Explorer
Description: Draw and explain the client-server architecture and trace what happens when you open a website.

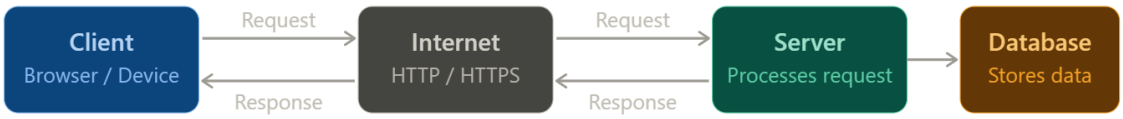
What is Client-Server Architecture?

Client-Server Architecture is a network model where two parties communicate over the internet — the Client requests something and the Server responds to it. This is the backbone of how every website works.

Components

1. Client - Your device (laptop/mobile) with a browser like Chrome that sends requests.
2. DNS Server - Converts the website name (like netflix.com) into an IP address the internet understands.
3. Server - A powerful computer that receives your request, processes it and sends back the website files.
4. Database - Stores all the data like user accounts, passwords, content etc. Server fetches from here if needed.
5. Internet - The medium through which everything communicates.

Client Server Architecture



What happens when you open netflix.com - Step by Step

Step 1 — You type netflix.com You enter the URL in Chrome and hit Enter.

Step 2 — DNS Lookup Your browser contacts the DNS Server. DNS converts netflix.com → 54.72.139.0 (IP address). Now the browser knows where to go.

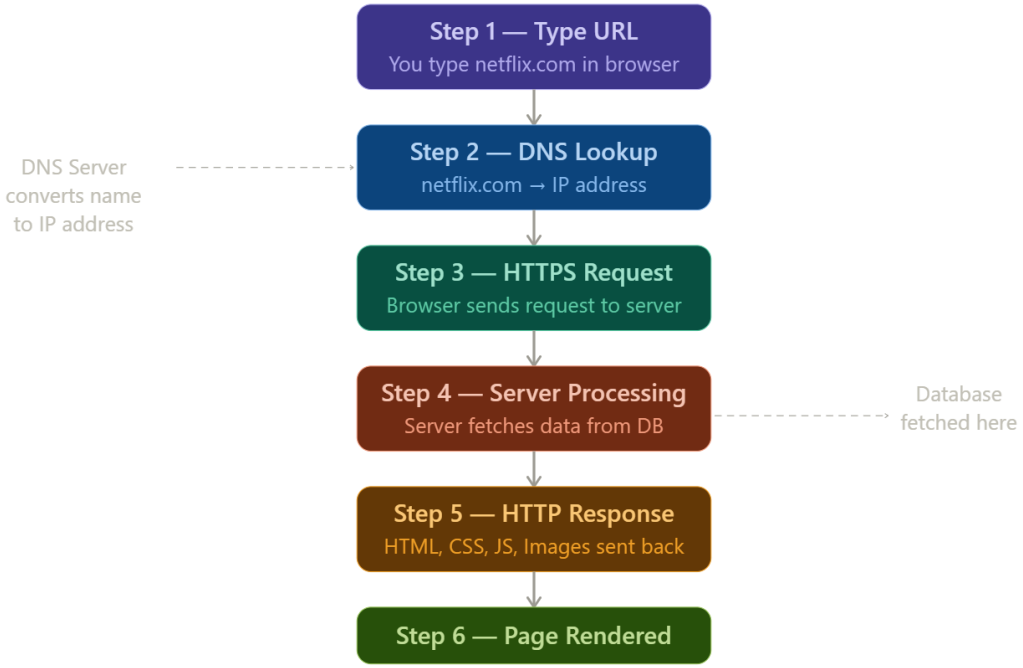
Step 3 — HTTP Request sent Browser sends an HTTPS request to Netflix's server at that IP address saying "give me the homepage."

Step 4 — Server processes the request Netflix's server receives your request, checks what you need, and fetches data from the database if required (like trending shows, your region etc.)

Step 5 — Server sends back a Response Server sends back HTML, CSS, JavaScript and image files to your browser.

Step 6 — Browser renders the page Your browser reads those files and displays the Netflix homepage on your screen.

Full Flow when you open netflix.com



Conclusion

Client-Server Architecture is the foundation of how the internet works. When we open a website like netflix.com, the browser first contacts a DNS server to get the IP address, then sends an HTTPS request to the server. The server processes the request, fetches data from the database if needed, and sends back HTML, CSS and JavaScript files. The browser then renders these files and displays the webpage. This entire process happens in milliseconds every time you visit a website.